

Statistics for AI - Exercise Exam Group 2 - SS2021

A committee of 5 people is to be selected from 10 men and 10 women.

- a) Calculate the probability that the committee will consist of 3 women and 2 men. [6 points]
- b) Calculate the probability that the committee will consist of at least 2 women. [6 points]
- c) Calculate the probability that the committee will consist of at most 1 man. [6 points]

a) 34.83 %

b) 84.83 %

c) 15.17 %

A continuous random variable X has the density $f(x) = k \cdot 0.5 + k \cdot x$ for $0 \leq x \leq 9$ (0 else).

- a) Calculate the constant k so that the probability density function is valid. [7 points]
- b) State the cumulative distribution function $F(x)$. [7 points]
- c) Calculate the probability of $P(1 \leq X \leq 4)$. [6 points]

a) $k = 1/45 = 0.222$

b) $F(x) = 1/45 \cdot 1/2 \cdot x + 1/45 \cdot 1/2 \cdot x^2 = x/90 + (x^2)/90$

c) 0.2

Three friends and seven other people are randomly seated in a row. What is the probability that all three friends will sit next to each other? [8 points]

A: $(8 \cdot 7! \cdot 3!) / 10! = 0.0667 = 6.67 \%$

The following fictitious data gives the results of an intelligence test in points and the reading time in seconds of a standardized text of 5 students.

Points on the intelligence test: 124 ; 126 ; 130 ; 119 ; 120

Reading time in seconds: 46 ; 44 ; 40 ; 42 ; 45

- a) Calculate an appropriate summary statistic to quantify the assumed linear relationship between the results of the intelligence test and the reading time. [15 points]
- b) Which reading time can be expected for a student who scored 115 points on this intelligence test? [15 points]

a)

mean $x = 123.8$

mean $y = 43.4$

variance $x = 16.16$

variance $y = 4.64$

covariance = -3.72

Bravais-Pearson Correlation Coefficient = $r_{xy} = -0.430$

b)

$y = a + b \cdot x$

$b = \text{covariance} / \text{variance}_x = -3.72 / 16.16 = -0.230198$

$a = \text{mean}_y - b \cdot \text{mean}_x = 43.4 - (-0.230198) \cdot 123.8 = 71.8985$

$y_i = a + b \cdot x_i = 71.8985 + (-0.230198) \cdot 115 = 45.426$

The length of salmon living in a specific river is normally distributed with an expected value of 71 cm and a variance of 36 cm². Now a bear catches a salmon at random.

- a) Calculate the probability that the length of this salmon is between 75 cm and 85 cm. [6 points]
- b) Calculate the probability that the length of this salmon is at most 71 cm. [6 points]
- c) Calculate the probability that the length of this salmon is at least 80 cm. [6 points]
- d) Calculate a symmetrical interval around the expected value in which the length of these salmon lays with a probability of 80%. [6 points]

- a) 0.243
- b) 0.5
- c) 0.067
- d) [63.311 ; 78.689]